



Statistics of Financial Markets

Seminar 7: Efficient markets, random walk and variance-ratio tests



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Acronyms used in this material (1/2)

- ▣ **ACF**: Autocorrelation Function
- ▣ **ADF**: Augmented Dickey–Fuller (test)
- ▣ **AI**: Artificial Intelligence
- ▣ **AIC**: Akaike Information Criterion
- ▣ **BET**: Bucharest Exchange Trading (index)
- ▣ **BIST**: Borsa İstanbul (BIST 100 index) — the Istanbul Stock Exchange (BIST 100 index)
- ▣ **BUX**: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- ▣ **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ **CD**: Chow–Denning (multiple variance-ratio test)
- ▣ **CDF**: Cumulative Distribution Function
- ▣ **COVID-19**: COronaVirus Disease 2019
- ▣ **EODHD**: EOD Historical Data (market-data provider)
- ▣ **FTSE**: Financial Times Stock Exchange (FTSE Russell, index provider)

Acronyms used in this material (2/2)

- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **KPSS**: Kwiatkowski–Phillips–Schmidt–Shin (stationarity test)
- ▣ **LB**: Ljung–Box (test)
- ▣ **OMV**: Österreichische Mineralölverwaltung — Austrian Mineral Oil Administration (OMV group)
- ▣ **PP**: Phillips–Perron (unit-root test)
- ▣ **PX**: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- ▣ **RW1**: Random Walk 1 (i.i.d. increments)
- ▣ **RW2**: Random Walk 2 (independent increments)
- ▣ **RW3**: Random Walk 3 (uncorrelated increments)
- ▣ **VR**: Variance Ratio
- ▣ **WIG20**: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Today's Question and Seminar Structure (1/4)

- Question of the seminar
 - ▶ **how can we test whether past prices predict future returns without fooling ourselves?**
- Three types of exercises
 - ▶ **A**: calculation exercises, on paper
 - ▶ **B**: exercises on real data, solved in the notebook
 - ▶ **C**: open questions, e.g. correcting an answer given by an AI tool
- Each exercise is marked
 - ▶ **[Solved]**: the full solution is presented in the seminar
 - ▶ **[Proposed]**: you solve it; the solution is discussed in the seminar

Today's Question and Seminar Structure (2/4)

- Today's exercises, in the order in which they are solved
- Part A: calculations on paper
 - ▶ A1: variance ratios from autocorrelations
 - ▶ A2: mean reversion in a volatile asset
 - ▶ A3: the Box–Pierce and Ljung–Box statistics, step by step
 - ▶ A4: Ljung–Box on squared returns
 - ▶ A5: the runs test on 20 days of signs
 - ▶ A6: a sequence with long runs
 - ▶ A7: ADF and KPSS decisions from regression output
 - ▶ A8: when ADF and KPSS disagree

Today's Question and Seminar Structure (3/4)

- Part B: real data, in the notebook
 - ▶ B1: autocorrelation and runs tests for the S&P 500
 - ▶ B2: the same tests for BET, Bitcoin and two BVB stocks
 - ▶ B3: unit root tests for the DAX
 - ▶ B4: unit root tests for BET, Bitcoin and TLV
 - ▶ B5: variance-ratio tests for the S&P 500
 - ▶ B6: variance-ratio tests for five emerging markets
- Part C: open questions
 - ▶ C1: has the BET become more efficient?
 - ▶ C2: audit an AI answer and correct its errors

Today's Question and Seminar Structure (4/4)

- Outputs to produce
 - ▶ the answers to the A exercises
 - ▶ the table or figure each B exercise asks for
 - ▶ a short argued answer to C1; the errors found in the AI answer of C2 and the corrected answer
 - ▶ the interpretation sentences each exercise asks for
- Nothing is handed in: the seminar is for practice; the solutions of the [Proposed] exercises are discussed in class
- Notebook for today: [open the seminar notebook in Google Colab](#)
 - ▶ each task names its notebook section
- Contact: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise Map

Task	Question	Type	Model
A1, A2	the square-root-of-time rule and variance ratios from autocorrelations	Solved, Proposed	A1
A3, A4	Box–Pierce and Ljung–Box statistics, step by step	Solved, Proposed	A3
A5, A6	the runs test on a sequence of signs	Solved, Proposed	A5
A7, A8	ADF and KPSS decisions from regression output	Solved, Proposed	A7
B1, B2	ACF, Ljung–Box and robust tests, runs test: S&P 500; BET, Bitcoin, BVB stocks	Solved, Proposed	B1
B3, B4	ADF, Phillips–Perron and KPSS on prices and returns: DAX; BET, Bitcoin, TLV	Solved, Proposed	B3
B5, B6	variance-ratio tests: S&P 500; five emerging markets	Solved, Proposed	B5
C1, C2	has the BET become more efficient? what is wrong in an AI answer?	Proposed	B1, B5

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data Used

Series	Source	Price used	Period
S&P 500, DAX	EODHD	close	1990–2026
BET	EODHD	close	1997–2026
Bitcoin	EODHD	close, 7 days a week	2014–2026
Banca Transilvania (TLV), OMV Petrom (SNP)	EODHD	adjusted close	2010–2026
WIG20, BUX, PX, BIST 100	EODHD	close	last ten years

- Daily log returns in %: $r_t = 100 (\ln P_t - \ln P_{t-1})$; weekends and repeated holiday closes are dropped (except Bitcoin); the last day is 18 September 2026
- TLV: the days 30 and 31 May 2016 are removed (a price adjustment applied one day late, [Chapter 2](#))
- In the notebook: `returns('sp500')`, `log_price('dax')`; no account or key is needed

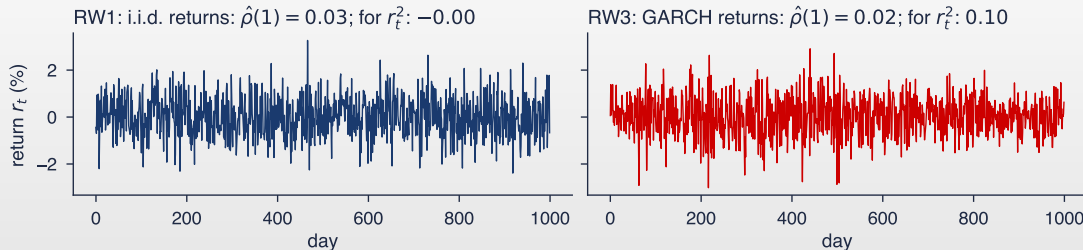
Prerequisites for Today (1/18): Weak-Form Efficiency

- **Weak-form efficiency Fama (1970):** past prices cannot be used to earn returns above the return that pays for risk
 - ▶ r_{t+1} : tomorrow's log return; μ : its equilibrium expected value; $r_{t+1} - \mu$: the excess return
 - ▶ excess returns are a **martingale difference**: $E[r_{t+1} - \mu \mid \text{past}] = 0$
 - ▶ in words: no information in past prices predicts the excess return
- The variance may still be predictable: volatility clustering is compatible with efficiency

Prerequisites for Today (2/18): Random Walks

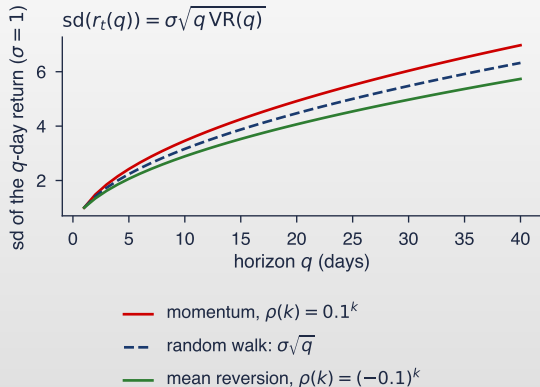
- Log price $p_t = \ln P_t$, **random walk**: $p_t = \mu + p_{t-1} + \varepsilon_t$, so $r_t = \mu + \varepsilon_t$ [Campbell, Lo & MacKinlay \(1997\)](#)
 - ▶ μ : the drift (mean daily return); ε_t : the shock of day t , mean 0, variance σ^2
- Three versions, from strong to weak
 - ▶ **RW1**: ε_t i.i.d. (independent, identically distributed); **RW2**: independent; **RW3**: uncorrelated
 - ▶ $\text{RW1} \Rightarrow \text{RW2} \Rightarrow \text{RW3}$; volatility clustering is allowed only by RW3 and by the martingale
- Under any of them, $\text{Var}(r_t(q)) = q\sigma^2$, with $r_t(q) = r_t + \dots + r_{t-q+1}$ the q -day return
 - ▶ “square-root-of-time rule”: $\text{sd}(r_t(q)) = \sigma\sqrt{q}$

Prerequisites for Today (3/18): RW1 and RW3 on a Chart



- Left: i.i.d. Normal returns (RW1); right: GARCH(1,1) returns, uncorrelated but with volatility clustering (RW3, not RW1)
- GARCH(1,1): $r_t = \sigma_t \varepsilon_t$, $\sigma_t^2 = \omega + a r_{t-1}^2 + b \sigma_{t-1}^2$, ε_t i.i.d. $N(0, 1)$
 - ▶ here $\omega = 0.02$, $a = 0.08$, $b = 0.9$: a large return raises the variance of the next day
 - ▶ both series have $\hat{\rho}(1) \approx 0$; only the squared returns tell them apart

Prerequisites for Today (4/18): the Square-Root-of-Time Rule



□ q -day return $r_t(q) = r_t + \dots + r_{t-q+1}$

$$\text{Var}(r_t(q)) = q\sigma^2 \text{VR}(q)$$

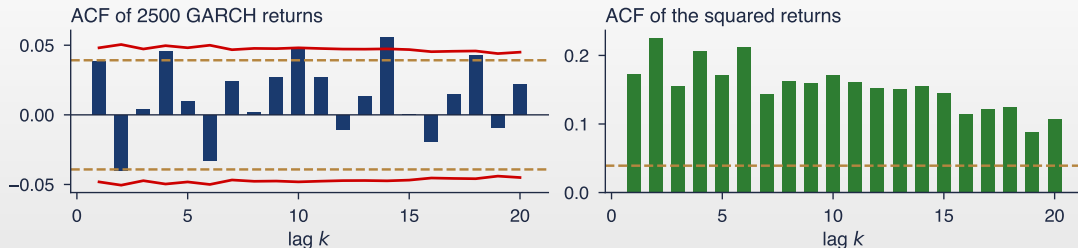
► $\text{VR}(q)$: the variance ratio, defined below; random walk: $\text{VR}(q) = 1$, so $sd = \sigma\sqrt{q}$ (dashed)

- Positive autocorrelations (red): the q -day risk is larger than $\sigma\sqrt{q}$
- Negative autocorrelations (green): it is smaller
- Chart: returns with $\rho(k) = \phi^k$, $\phi = 0.1$ and $\phi = -0.1$

Prerequisites for Today (5/18): the Sample ACF

- Sample **ACF** (autocorrelation function): $\hat{\rho}(k) = \sum_{t \geq k} e_t e_{t-k} / \sum_t e_t^2$
 - ▶ $e_t = r_t - \bar{r}$: the demeaned return; k : the lag in days; T : the number of returns
 - ▶ $\hat{\rho}(k)$: the correlation between today's return and the return k days earlier
- i.i.d. returns: $\text{Var}(\hat{\rho}(k)) \approx 1/T$
 - ▶ 95% band $\pm 1.96/\sqrt{T}$: values outside it are significant at 5%
- Robust (volatility clustering): $\text{Var}(\hat{\rho}(k)) \approx \hat{\delta}(k)/T$ [Lo & MacKinlay \(1988\)](#)
 - ▶ $\hat{\delta}(k) = T \sum_t e_t^2 e_{t-k}^2 / (\sum_t e_t^2)^2$: about 1 for i.i.d. returns, above 1 when large returns follow large returns

Prerequisites for Today (6/18): i.i.d. and Robust Bands on a Chart

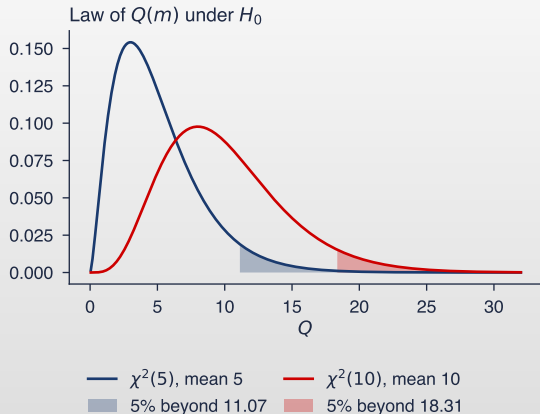


— robust band $\pm 1.96\sqrt{\hat{\delta}(k)/T}$ - - - i.i.d. band $\pm 1.96/\sqrt{T}$ ■ $\hat{\rho}(k)$ ■ ACF of r_t^2

- Left: the ACF of 2500 simulated GARCH returns, uncorrelated by construction
 - ▶ the i.i.d. band (dashed) is too narrow: some $\hat{\rho}(k)$ fall outside it by chance
 - ▶ the robust band (red) is wider: here $\hat{\delta}(1) = 1.51$
- Right: the squared returns are strongly autocorrelated: volatility clustering, which makes $\hat{\delta}(k) > 1$

Prerequisites for Today (7/18): Portmanteau Tests

- $H_0: \rho(1) = \dots = \rho(m) = 0$: no autocorrelation in the first m lags
 - ▶ three statistics, each $\approx \chi^2(m)$ (chi-square with m degrees of freedom) under its null
- Box–Pierce $Q_{BP} = T \sum_{k=1}^m \hat{\rho}(k)^2$ Box & Pierce (1970)
 - ▶ LB (Ljung–Box) $Q_{LB} = T(T+2) \sum_{k=1}^m \hat{\rho}(k)^2 / (T-k)$ Ljung & Box (1978): the same idea, closer to χ^2 in small samples
- Robust $\tilde{Q} = \sum_{k=1}^m T \hat{\rho}(k)^2 / \hat{\delta}(k)$ Escanciano & Lobato (2009)
 - ▶ each squared autocorrelation divided by its own variance: valid for a martingale with volatility clustering
- 5% critical values: $\chi_{0.95}^2(3) = 7.81$, $\chi_{0.95}^2(5) = 11.07$, $\chi_{0.95}^2(10) = 18.31$

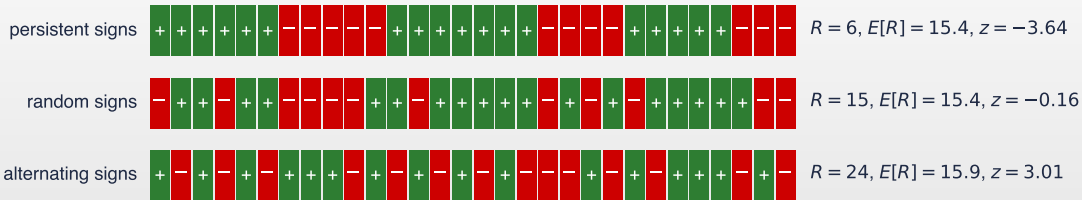
Prerequisites for Today (8/18): the Chi-Square Law of Q 

- Under H_0 , Q_{BP} , Q_{LB} and $\tilde{Q}$ follow approximately $\chi^2(m)$
 - ▶ each $T\hat{\rho}(k)^2$ is roughly a squared $N(0, 1)$ variable; their sum over m lags is $\chi^2(m)$, with mean m
- Reject H_0 when Q falls in the shaded 5% tail: beyond 11.07 ($m = 5$) or 18.31 ($m = 10$)
- p-value: the area to the right of the observed Q
- A single large $\hat{\rho}(k)$ can be enough to push Q into the tail

Prerequisites for Today (9/18): the Runs Test

- A **run**: a maximal sequence of returns with the same sign [Wald & Wolfowitz \(1940\)](#); zero returns are dropped
 - ▶ $++---+-$: $R = 4$ runs; $n_1 = 3$ positive, $n_2 = 4$ negative, $n = n_1 + n_2 = 7$
- Under independence of the signs:
 - ▶ $E[R] = \frac{2n_1n_2}{n} + 1$, $\text{Var}(R) = \frac{2n_1n_2(2n_1n_2 - n)}{n^2(n - 1)}$
 - ▶ $z = (R - E[R]) / \sqrt{\text{Var}(R)} \approx N(0, 1)$; reject at 5% if $|z| > 1.96$
- Reading z
 - ▶ $z < 0$: too few runs, signs persist (positive dependence, trends)
 - ▶ $z > 0$: too many runs, signs alternate (negative dependence, reversals)
- Signs only: the test ignores the size of the returns, so heavy tails and volatility do not affect it

Prerequisites for Today (10/18): Runs on a Chart

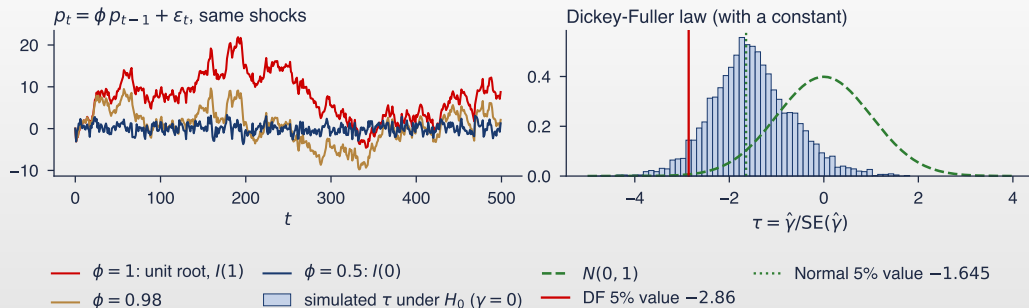


- Three sequences of 30 signs (green +, red -); R counts the blocks of equal signs
- Persistent signs: few long runs, $z < -1.96$: positive dependence (trends)
- Random signs: R close to $E[R]$, $|z| < 1.96$: independence is not rejected
- Alternating signs: many short runs, $z > 1.96$: negative dependence (reversals)

Prerequisites for Today (11/18): Unit Roots

- $I(0)$: stationary (constant mean and variance, shocks fade)
 - ▶ $I(1)$: the first difference $\Delta p_t = p_t - p_{t-1}$ is $I(0)$; a random walk is $I(1)$
- Dickey–Fuller regression [Dickey & Fuller \(1979\)](#): $\Delta p_t = c [+ bt] + \gamma p_{t-1} [+ \sum_j \delta_j \Delta p_{t-j}] + \varepsilon_t$
 - ▶ c : constant; bt : optional trend; δ_j : coefficients of the optional lagged differences
 - ▶ $\gamma = 0$: a unit root (shocks are permanent); $\gamma < 0$: the series returns to a level
 - ▶ H_0 : $\gamma = 0$; $\tau = \hat{\gamma}/\text{SE}(\hat{\gamma})$; reject if τ is below the critical value
- 5% critical values [MacKinnon \(1996\)](#): -2.86 (constant), -3.41 (constant and trend); not -1.645

Prerequisites for Today (12/18): Unit Roots on a Chart



- Left: the same shocks with $\phi = 1$ (a random walk: shocks never fade), $\phi = 0.98$ (they fade slowly) and $\phi = 0.5$ (fast)
 - in the Dickey-Fuller regression $\gamma = \phi - 1$: $\gamma = 0$ is the unit root
- Right: 4 000 simulations of τ under H_0 : the law lies to the left of $N(0, 1)$
 - simulated 5% quantile -2.84 , close to -2.86 ; with -1.645 a true unit root would be rejected far more often than 5%

Prerequisites for Today (13/18): ADF, PP and KPSS

- **ADF** (augmented Dickey–Fuller): the regression with the lagged differences [Said & Dickey \(1984\)](#)
 - ▶ **PP** (Phillips–Perron): without them, with a corrected τ [Phillips & Perron \(1988\)](#); same H_0 and critical values
- **KPSS** [Kwiatkowski et al. \(1992\)](#): the hypotheses are reversed, H_0 stationarity
 - ▶ reject if the statistic exceeds 0.463 (constant) or 0.146 (trend)
- Combining the two
 - ▶ ADF does not reject + KPSS rejects: $I(1)$; ADF rejects + KPSS does not: $I(0)$; otherwise: inconclusive

Prerequisites for Today (14/18): the KPSS Statistic and the Lag Choice

- **KPSS:** e_t = residuals of p_t on a constant (and a trend)

$$\text{KPSS} = \frac{1}{T^2 \hat{\sigma}_{LR}^2} \sum_{t=1}^T S_t^2, \quad S_t = e_1 + \dots + e_t$$

- ▶ S_t : the partial sums; with a unit root they wander far from 0, so KPSS is large

- Long-run variance with l Bartlett lags (Newey–West)

$$\hat{\sigma}_{LR}^2 = \hat{\gamma}_0 + 2 \sum_{j=1}^l \left(1 - \frac{j}{l+1}\right) \hat{\gamma}_j$$

- ▶ $\hat{\gamma}_j = \frac{1}{T} \sum_{t>j} e_t e_{t-j}$: the autocovariance at lag j ; PP corrects τ with the same kind of estimate

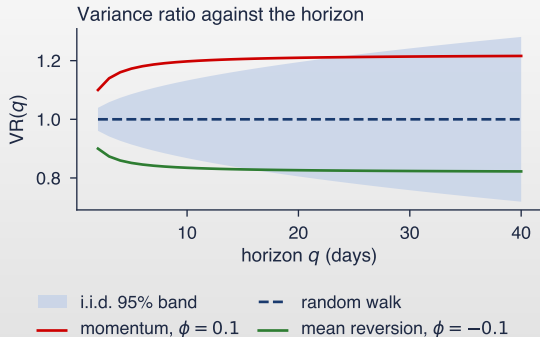
- ADF lag order by AIC (Akaike information criterion)

- ▶ $\text{AIC} = T \ln \hat{\sigma}_\varepsilon^2 + 2k$ for each candidate number of lags; keep the lowest
- ▶ $\hat{\sigma}_\varepsilon^2$: the residual variance; k : the number of estimated coefficients

Prerequisites for Today (15/18): Variance Ratios

- ▣ **VR** (variance ratio): $VR(q) = \frac{\text{Var}(r_t(q))}{q \text{Var}(r_t)} = 1 + 2 \sum_{k=1}^{q-1} \left(1 - \frac{k}{q}\right) \rho(k)$ Lo & MacKinlay (1988)
 - ▶ the q -day variance relative to q daily variances; $= 1$: random walk; > 1 : momentum; < 1 : mean reversion
- ▣ Tests with T returns, both $\approx N(0, 1)$ under H_0
 - ▶ i.i.d.: $Z(q) = (\widehat{VR}(q) - 1) / \sqrt{2(2q - 1)(q - 1) / (3qT)}$
 - ▶ robust: $Z^*(q) = (\widehat{VR}(q) - 1) / \sqrt{\hat{\theta}(q) / T}$, $\hat{\theta}(q) = \sum_{j=1}^{q-1} [2(q - j) / q]^2 \hat{\delta}(j)$
 - ▶ $\widehat{VR}(q)$: the estimate from the data; $\hat{\delta}(j)$: as in the robust ACF band (5/18)

Prerequisites for Today (16/18): Variance Ratios on a Chart

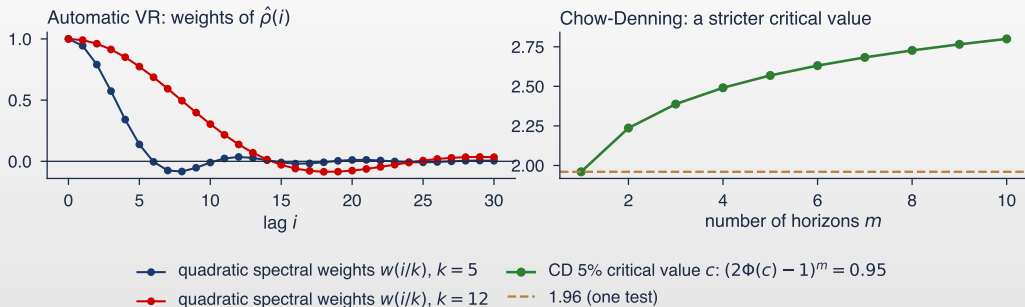


- VR(q) for returns with $\rho(k) = \phi^k$
 - ▶ $\phi = 0.1$ (red): above 1, momentum; $\phi = -0.1$ (green): below 1, mean reversion
- Shaded: the i.i.d. 95% band
 $1 \pm 1.96 \sqrt{2(2q-1)(q-1)/(3qT)}$, $T = 2500$
 - ▶ it widens with q : long horizons are measured less precisely
 - ▶ here $\phi = \pm 0.1$ is detected at short horizons, but not at long ones

Prerequisites for Today (17/18): Several Horizons and Several Markets

- Several horizons: **CD** (Chow–Denning) = $\max_i |Z^*(q_i)|$, $i = 1, \dots, m$ [Chow & Denning \(1993\)](#)
 - ▶ critical value c from $P(\text{CD} \leq c) = (2\Phi(c) - 1)^m$, with Φ the $N(0, 1)$ CDF; 5%, $m = 4$: $c = 2.49$
- Several markets (Šidák): each of m tests at level $1 - 0.95^{1/m}$
 - ▶ the probability of at least one false rejection then stays at 5%
- Automatic VR [Choi \(1999\)](#): $\widehat{\text{VR}}(k) = 1 + 2 \sum_{i \geq 1} w(i/k) \hat{\rho}(i)$
 - ▶ w : weights that decrease smoothly; the bandwidth k is chosen from the data
 - ▶ p-value by wild bootstrap [Kim \(2009\)](#): $r_t^* = \eta_t(r_t - \bar{r})$, $\eta_t \sim N(0, 1)$, which keeps the volatility and removes any autocorrelation

Prerequisites for Today (18/18): Automatic VR and Chow–Denning on a Chart



- Left: the automatic VR weights each $\hat{\rho}(i)$ by $w(i/k)$: close to 1 for small lags, near 0 beyond about $1.2k$
 - ▶ quadratic spectral kernel; a larger bandwidth k uses more lags
- Right: the Chow–Denning critical value c for m horizons
 - ▶ $m = 1$: 1.96; $m = 4$: 2.49; it grows with m so that all m tests together keep a 5% error

A1: Variance Ratios from Autocorrelations [Solved]

Task

- ▣ Daily returns have a standard deviation $\sigma = 1.2\%$.
- ▣ 1. Under RW1, compute the standard deviation of the 5-day and of the 20-day return.
- ▣ 2. Suppose $\rho(1) = 0.10$, $\rho(2) = 0.05$ and $\rho(k) = 0$ for $k \geq 3$; compute VR(2) and VR(5).
- ▣ 3. Treat these as estimates from $T = 2500$ i.i.d. returns and compute $Z(2)$ and $Z(5)$.
- ▣ 4. Compute the 5-day standard deviation implied by VR(5).
- ▣ Report: two standard deviations, two VR, two Z and one sentence.

Solution

- ▣ 1. $1.2\sqrt{5} = 2.68\%$; $1.2\sqrt{20} = 5.37\%$
- ▣ 2. $VR(2) = 1 + 0.10 = 1.10$; $VR(5) = 1 + 2(0.8 \times 0.10 + 0.6 \times 0.05) = 1.220$
- ▣ 3. $\sqrt{2 \cdot 3 \cdot 1 / (6 \cdot 2500)} = 0.020$, $Z(2) = 5.00$;
 $\sqrt{2 \cdot 9 \cdot 4 / (15 \cdot 2500)} = 0.0438$, $Z(5) = 5.02$
- ▣ 4. $1.2\sqrt{5} \times 1.220 = 2.96\%$ instead of 2.68%: momentum makes weekly risk larger than the $\sqrt{q}$ rule says.

A2: Mean Reversion in a Volatile Asset [Proposed]

Solution discussed in class

Task

- Daily returns: $\sigma = 2\%$, $\rho(1) = -0.08$, $\rho(2) = 0.03$, $\rho(k) = 0$ for $k \geq 3$, $T = 1000$. Model: A1.
- 1. Compute $VR(2)$ and $VR(5)$.
- 2. Compute $Z(2)$ and $Z(5)$ and decide at 5%.
- 3. Compare the 5-day standard deviation implied by $VR(5)$ with $2\sqrt{5}$.
- Report: two VR, two Z, two standard deviations and one sentence.

A3: Box–Pierce and Ljung–Box, Step by Step [Solved]

Task

- ▣ $T = 500$ daily returns;
 $\hat{\rho}(1), \dots, \hat{\rho}(5) = 0.12, -0.05, 0.04, 0.03, -0.06$.
- ▣ 1. Compute $Q_{BP}(5)$.
- ▣ 2. Compute $Q_{LB}(5)$.
- ▣ 3. Compare with $\chi^2_{0.95}(5) = 11.07$ and decide.
- ▣ 4. Say which lag drives the result.
- ▣ Report: two statistics, a decision and one sentence.

Solution

- ▣ 1. $T\hat{\rho}(k)^2$: 7.20, 1.25, 0.80, 0.45, 1.80; $Q_{BP}(5) = 11.50$
- ▣ 2. $500 \cdot 502 \hat{\rho}(k)^2 / (500 - k)$: 7.24, 1.26, 0.81, 0.46, 1.83;
 $Q_{LB}(5) = 11.59$
- ▣ 3. $11.59 > 11.07$: reject H_0 at 5% ($p = 0.041$)
- ▣ 4. Lag 1 alone gives 7.20 of 11.50: one autocorrelation of 0.12 drives the rejection.

A4: Ljung–Box on Squared Returns [Proposed]

Solution discussed in class

Task

- ▣ $T = 750$; the ACF of the squared returns: 0.25, 0.20, 0.18, 0.15, 0.12 at lags 1–5. Model: A3.
- ▣ 1. Compute $Q_{LB}(5)$ and decide at 5%.
- ▣ 2. Say which of RW1, RW3 and the martingale this rejects.
- ▣ 3. Say whether weak-form efficiency is rejected.
- ▣ Report: one statistic and two sentences.



A5: The Runs Test on 20 Days [Solved]

Task

- ▣ Signs of 20 daily returns:
++-++++--+- - -++-+- -++.
- ▣ 1. Count n_1 , n_2 and the number of runs R .
- ▣ 2. Compute $E[R]$ and $\text{Var}(R)$.
- ▣ 3. Compute z and decide at 5%.
- ▣ Report: R , $E[R]$, z and one sentence.

Solution

- ▣ 1. $n_1 = 11$, $n_2 = 9$, $R = 11$
- ▣ 2. $E[R] = 2 \cdot 11 \cdot 9/20 + 1 = 10.90$;
 $\text{Var}(R) = 198 \cdot 178/(400 \cdot 19) = 4.64$
- ▣ 3. $z = (11 - 10.90)/2.15 = 0.05$ ($p = 0.963$): do not reject independence.
- ▣ With 20 days the test has little power: only large departures can be detected.

A6: A Sequence with Long Runs [Proposed]

Solution discussed in class

Task

- ▣ Signs of 24 daily returns: $+++---++++---++++---++--$. Model: A5.
- ▣ 1. Count n_1 , n_2 and R .
- ▣ 2. Compute $E[R]$, $\text{Var}(R)$ and z .
- ▣ 3. Decide at 5% and say what kind of dependence the signs show.
- ▣ Report: R , $E[R]$, z and one sentence.



A7: ADF and KPSS from Regression Output [Solved]

Task

- ▣ Log price, $T = 2500$:
 $\Delta p_t = 0.021 - 0.0028 p_{t-1} + 0.05 \Delta p_{t-1} + e_t$,
 $SE(\hat{\gamma}) = 0.0019$; KPSS (constant) = 2.10.
- ▣ Returns: $\Delta r_t = 0.03 - 0.92 r_{t-1} + e_t$, $SE(\hat{\gamma}) = 0.020$; KPSS (constant) = 0.09.
- ▣ 1. Compute τ for the price and for the returns.
- ▣ 2. Decide with the 5% critical value for a constant.
- ▣ 3. Decide with KPSS and give the order of integration of each series.
- ▣ Report: two τ , four decisions and one sentence.

Solution

- ▣ 1. Price: $\tau = -0.0028/0.0019 = -1.47$; returns:
 $\tau = -0.92/0.020 = -46.0$
- ▣ 2. Price: $-1.47 > -2.86$: do not reject a unit root; returns:
 $-46.0 < -2.86$: reject
- ▣ 3. KPSS: price $2.10 > 0.463$: reject stationarity; returns
 $0.09 < 0.463$: do not reject
- ▣ The price is $I(1)$, the returns $I(0)$: the picture of a random walk, but not yet a proof of it.

A8: When the Tests Disagree [Proposed]

Solution discussed in class

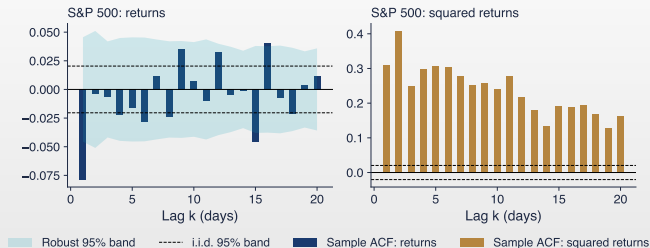
Task

- ▣ Log price of a stock, $T = 1500$, constant and trend: $\hat{\gamma} = -0.0105$, $SE(\hat{\gamma}) = 0.0029$; KPSS (trend) = 0.21. Model: A7.
- ▣ 1. Compute τ and decide with the 5% critical value for a constant and a trend.
- ▣ 2. Decide with KPSS.
- ▣ 3. Combine the two decisions and give two possible reasons for the conflict.
- ▣ Report: τ , two decisions and two sentences.

B1: Autocorrelation in the S&P 500 [Solved]

- **Question:** does the verdict on the autocorrelation of daily S&P 500 returns depend on whether we allow for volatility clustering?
- **Inputs:** S&P 500 closes since 1990, daily log returns in %
- **Tasks**
 1. Draw the ACF of the returns (lags 1–20) with the i.i.d. and the robust 95% bands, and the ACF of the squared returns.
 2. Report $\hat{\rho}(1), \dots, \hat{\rho}(5)$ of the returns and of the squared returns.
 3. Compute $Q_{LB}(10)$ and the robust $\tilde{Q}(10)$, with p-values, for the returns and for the squared returns.
 4. Say which random walk hypothesis each test rejects.
 5. Interpretation: is weak-form efficiency rejected?
- **Report:** the chart, ten autocorrelations, four statistics with p-values and two sentences
- **Notebook:** section [B1](#)

B1: Solution [Solved]



- ▣ $\hat{\rho}(1), \dots, \hat{\rho}(5)$, returns: $-0.079, -0.004, -0.006, -0.022, -0.016$; squared: $0.311, 0.408, 0.250, 0.298, 0.308$; $T = 9\,240$
- ▣ Returns: $Q_{LB}(10) = 90.8$ ($p < 0.001$), $\tilde{Q}(10) = 18.8$ ($p = 0.043$); squared: $Q_{LB}(10) = 8015$, $\tilde{Q}(10) = 189.4$ (both $p < 0.001$)
- ▣ Interpretation: squared returns reject RW1 and RW2 (volatility clustering)
 - the robust test rejects RW3 only barely, because of a small negative $\hat{\rho}(1)$: a weak, short-lived reversal

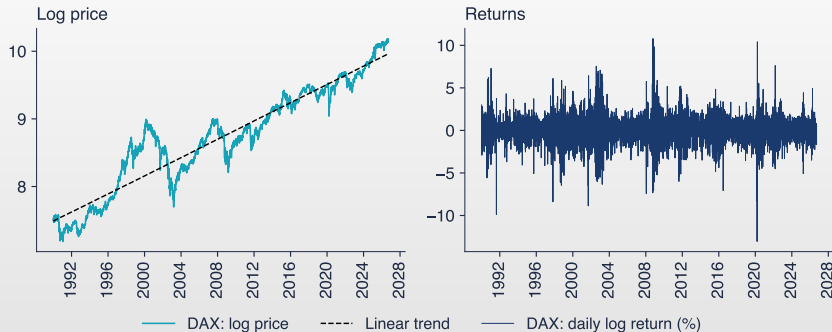
B2: BET, Bitcoin and Two BVB Stocks [Proposed]

- **Question:** for which of the BET, Bitcoin, Banca Transilvania and OMV Petrom does the verdict on autocorrelation change between the classic and the robust test? Model: [B1](#).
- **Inputs:** BET since 1997, Bitcoin since 2014, TLV and SNP since 2010 (TLV without 30–31 May 2016), daily log returns in %
- **Tasks**
 1. Compute $\hat{\rho}(1)$, $Q_{LB}(10)$ and $\tilde{Q}(10)$ with p-values for each series.
 2. Compute the runs test z and its p-value for each series.
 3. Mark the series for which the classic and the robust test disagree at 5%.
 4. Interpretation: why is the BET different from the other three?
- **Report:** one table and two sentences
- **Notebook:** section [B2](#)

B3: Unit Roots in the DAX [Solved]

- **Question:** what is the order of integration of the DAX log price and of its returns?
- **Inputs:** DAX closes since 1990; log price and daily log returns
- **Tasks**
 1. Draw the log price with a fitted linear trend, and the returns.
 2. Run ADF (lag order by AIC), PP and KPSS on the log price with a constant and a trend.
 3. Run the same three tests on the returns with a constant.
 4. Interpretation: does a unit root in the DAX price prove that the DAX is weak-form efficient?
- **Report:** the chart, six statistics with p-values and two sentences
- **Notebook:** section [B3](#)

B3: Solution (1/2) [Solved]



□ Left: the DAX log price with a fitted linear trend; right: the daily log returns

B3: Solution (2/2) [Solved]

	ADF	PP	KPSS
log price (trend)	-2.69 ($p = 0.239$)	-2.65 ($p = 0.260$)	0.94 ($p < 0.01$)
returns (constant)	-23.30 ($p < 0.001$)	-97.40 ($p < 0.001$)	0.04 ($p > 0.10$)

- ▣ ADF used 30 lags for the price (AIC); PP and KPSS use 38 Bartlett lags; $T = 9\,288$
- ▣ Price: no rejection by ADF or PP, rejection by KPSS: $I(1)$; returns: the opposite: $I(0)$
- ▣ Interpretation: no; a unit root is necessary for a random walk but allows predictable increments; efficiency is tested on the returns (B1, B5)

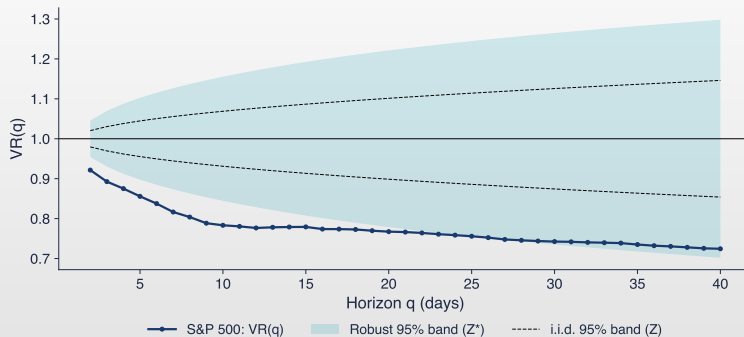
B4: BET, Bitcoin and TLV [Proposed]

- **Question:** do the three unit-root tests agree for the BET, Bitcoin and Banca Transilvania? Model: [B3](#).
- **Inputs:** BET since 1997, Bitcoin since 2014, TLV since 2010 (without 30–31 May 2016)
- **Tasks**
 1. Run ADF, PP and KPSS on each log price (constant and trend) and on each return series (constant).
 2. Classify each series as $I(1)$, $I(0)$ or inconclusive.
 3. For an inconclusive case, propose one check that could settle it.
 4. Interpretation: would you trade a “mean-reverting” TLV price on this evidence?
- **Report:** one table, three classifications and two sentences
- **Notebook:** section [B4](#)

B5: Variance Ratios of the S&P 500 [Solved]

- **Question:** is the S&P 500 mean-reverting over horizons of 2 to 20 days, once volatility clustering is taken into account?
- **Inputs:** S&P 500 closes since 1990, daily log returns in %
- **Tasks**
 1. Compute $VR(q)$, $Z(q)$ and $Z^*(q)$ for $q = 2, 5, 10, 20$.
 2. Compute the Chow–Denning statistic and compare it with its 5% critical value.
 3. Draw $VR(q)$ for $q = 2, \dots, 40$ with the i.i.d. and the robust 95% bands.
 4. Interpretation: is the mean reversion large enough to matter for an investor?
- **Report:** a table of twelve numbers, one statistic, the chart and two sentences
- **Notebook:** section [B5](#)

B5: Solution (1/2) [Solved]



□ VR(q) of the S&P 500 for $q = 2, \dots, 40$, with the i.i.d. and the robust 95% bands

B5: Solution (2/2) [Solved]

	$q = 2$	$q = 5$	$q = 10$	$q = 20$
$VR(q)$	0.922	0.856	0.783	0.767
$Z(q)$	-7.54	-6.33	-6.17	-4.50
$Z^*(q)$	-3.37	-2.76	-2.73	-2.06

- $T = 9\,240$; Chow–Denning: $\max |Z^*| = 3.37 > 2.49$ ($p = 0.003$): reject the random walk
- $Z(2)$ is about 2.2 times $Z^*(2)$: ignoring volatility clustering exaggerates the evidence
- Interpretation: significant, but small
 - ▶ the weekly variance is about 86% of five daily variances
 - ▶ after trading costs this is hardly a profit

B6: Five Emerging Markets [Proposed]

- **Question:** which of the WIG20, BUX, PX, BET and BIST 100 departs from a random walk over the last ten years? Model: B5.
- **Inputs:** daily log returns from 19 September 2016 to 18 September 2026
- **Tasks**
 1. Compute $VR(5)$, $Z^*(5)$ and its p-value for each market.
 2. Compute the Chow–Denning p-value ($q = 2, 5, 10, 20$) and the automatic VR test with its wild-bootstrap p-value.
 3. Compute the Šidák critical value for five markets tested together at 5%.
 4. Interpretation: which markets would you call inefficient once you account for testing five markets?
- **Report:** one table, one critical value and two sentences
- **Notebook:** section B6

C1: Has the BET Become More Efficient? [Proposed]

- **Question:** how has the predictability of the BET evolved since 1997, including around the FTSE Russell upgrade of September 2020?
- **Inputs:** BET daily log returns since 1997; models: [B1](#), [B5](#)
- **Tasks**
 1. Compute $VR(5)$ and $Z^*(5)$ on rolling windows of 500 days, one every 21 days, and the share of windows with $|Z^*(5)| > 1.96$ in the first and in the last five years.
 2. Compute $\hat{\rho}(1)$, $VR(5)$, $Z^*(5)$ and the Chow–Denning p-value in 1997–2006, 2007–2016 and 2017–2026.
 3. Compare $VR(5)$ in the five years before and after 21 September 2020.
 4. List what else changed in the Romanian market at the same time.
 5. Interpretation: how would you separate these changes from the effect of the upgrade?
- **Report:** a table, one chart and a plan for a project
- **Notebook:** section [C1](#)

C2: Audit an AI Answer [Proposed]

- A student asked an AI assistant whether the DAX (daily, since 1990) is weak-form efficient. The answer:
- (a) ADF on the DAX log price gives $\tau = -2.69$, $p = 0.24$: the unit root is not rejected, so the DAX is weak-form efficient.
- (b) Ljung-Box $Q(10) = 21.83$, $p = 0.016$: DAX returns are predictable, so the DAX is not efficient.
- (c) For squared returns $Q(10) = 3747$: DAX returns are not a martingale.
- (d) $VR(10) = 0.93 < 1$ shows positive autocorrelation (momentum) in the DAX.
- (e) Under RW1, $VR(q) = 1$ at every horizon and the variance of q -day returns is q times the daily variance.
- **Tasks**
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
 - ▶ 2. Report: a list of five verdicts with one line of justification each.

What You Should Take from Today

- ▣ A unit root in prices is the starting point, not the answer: efficiency is about the returns
- ▣ Test the martingale with robust statistics ($\tilde{Q}, Z^*$): classic tests mistake volatility clustering for predictability
- ▣ $VR > 1$: momentum; $VR < 1$: mean reversion; the S&P 500 reverts slightly, the old BET trended
- ▣ Many horizons or many markets: use Chow–Denning or a Šidák correction
- ▣ An AI answer is a draft: check the null of each test, the sign of $VR - 1$ and the robust version

After the Seminar

- Lecture 7 develops each topic of today: efficiency, random walks, ACF tests, unit roots, variance ratios, adaptive markets
- Try the [Proposed] tasks in the notebook; the solutions are discussed in class
- C1 can grow into a team project: BET stocks, WIG20 and BUX as controls, a block bootstrap of the difference
- Reading: [Franke, Härdle & Hafner \(2019\)](#), Ch. 11; exercises in [Borak, Härdle & López-Cabrera \(2013\)](#), Ch. 11; [Campbell, Lo & MacKinlay \(1997\)](#), Ch. 2

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